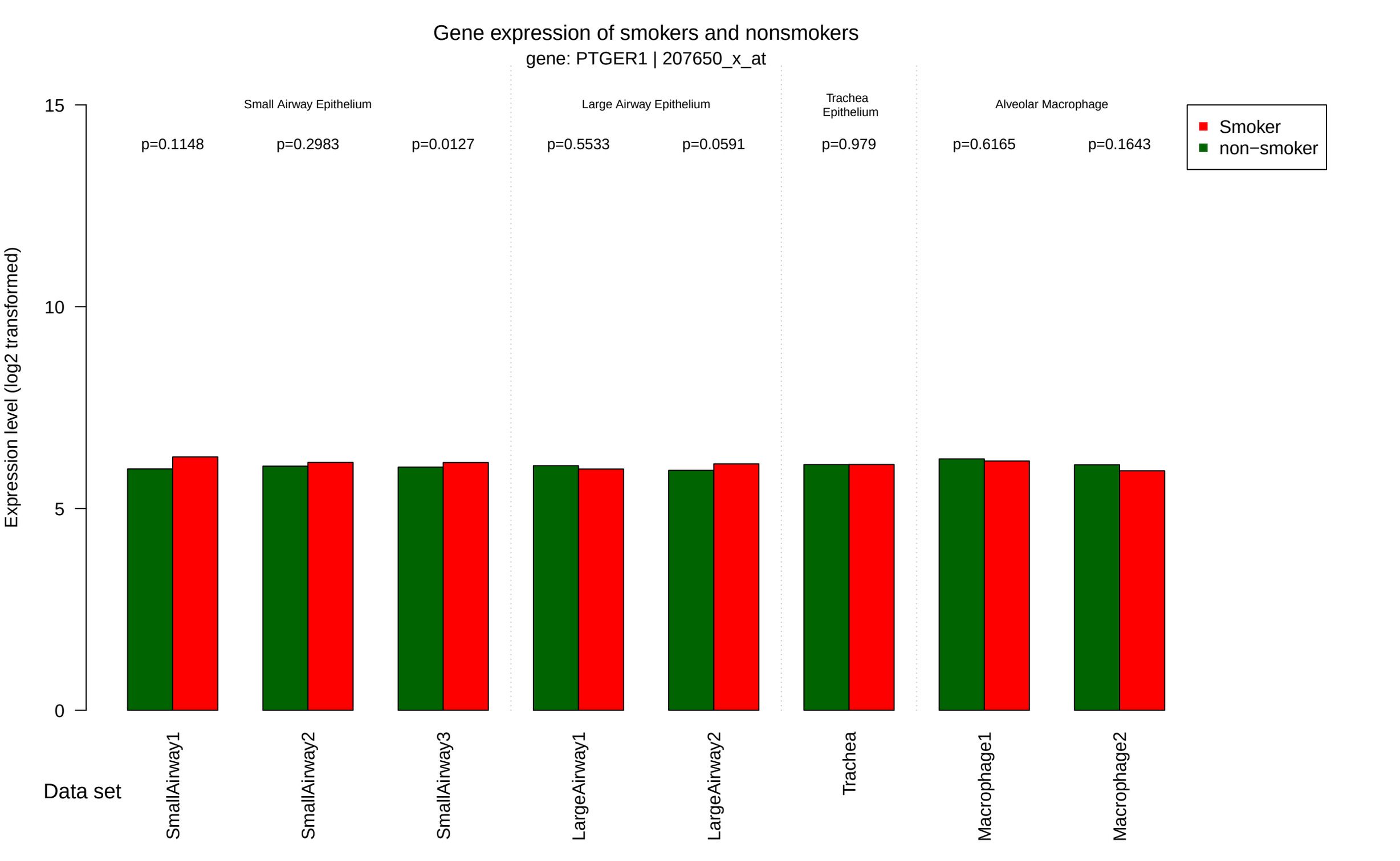
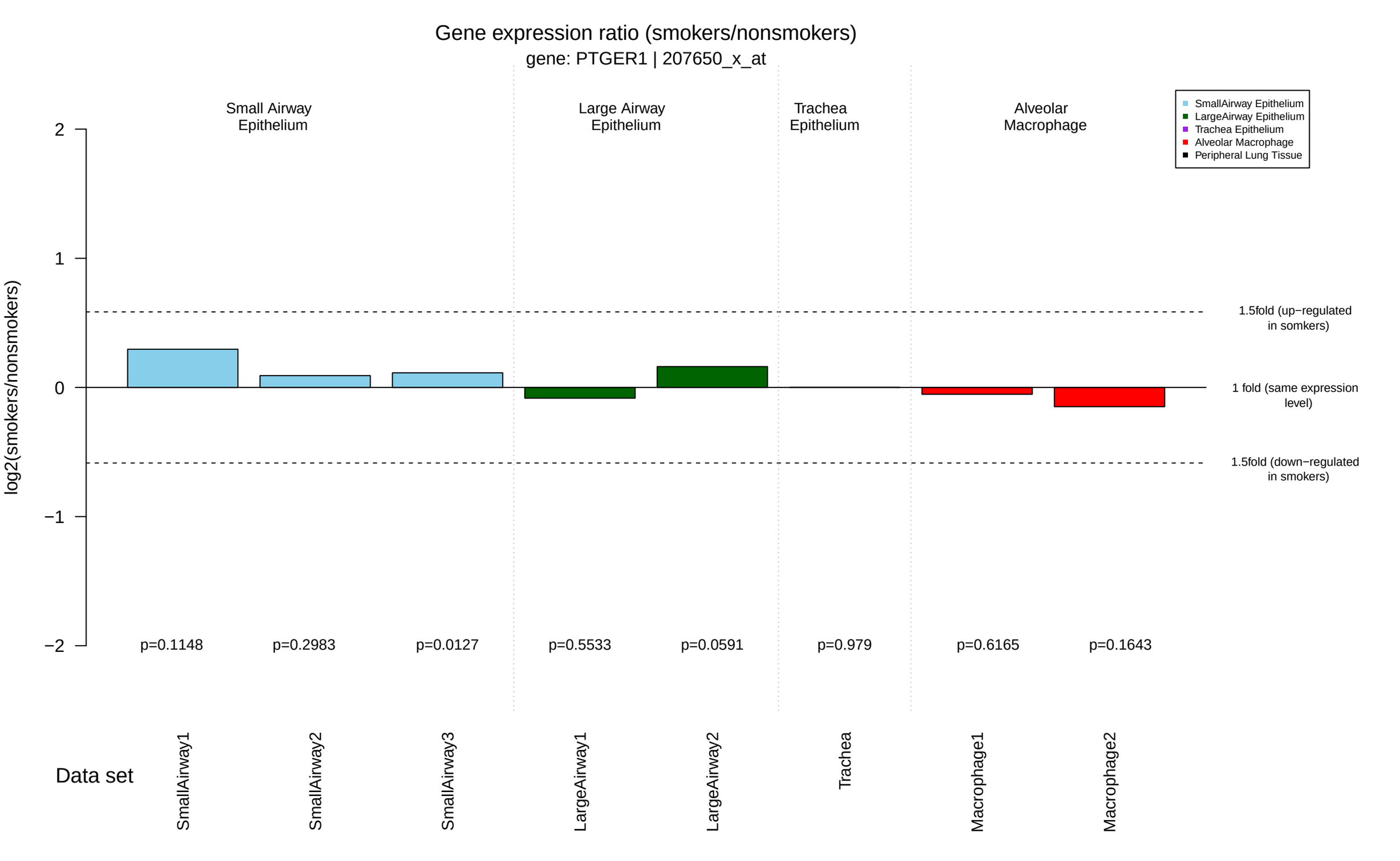
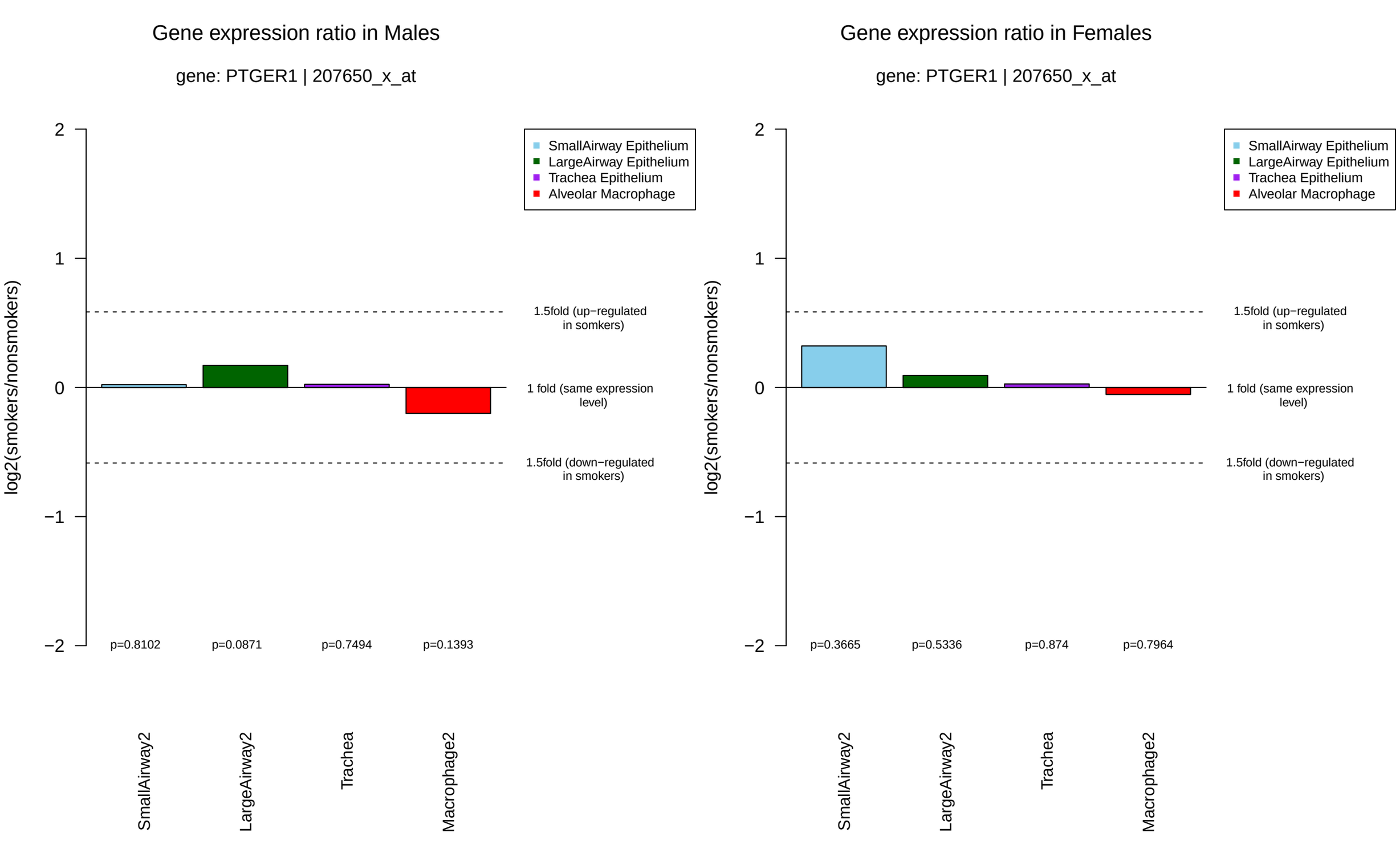
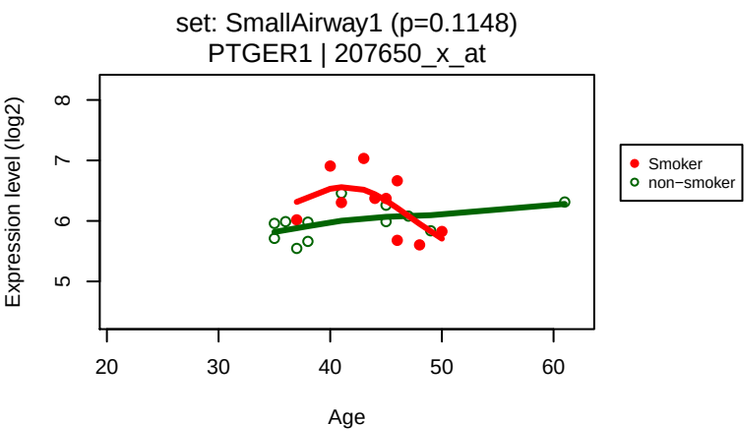
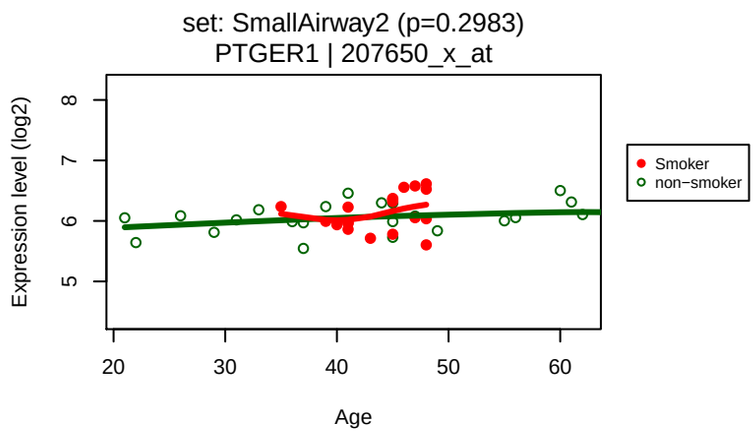
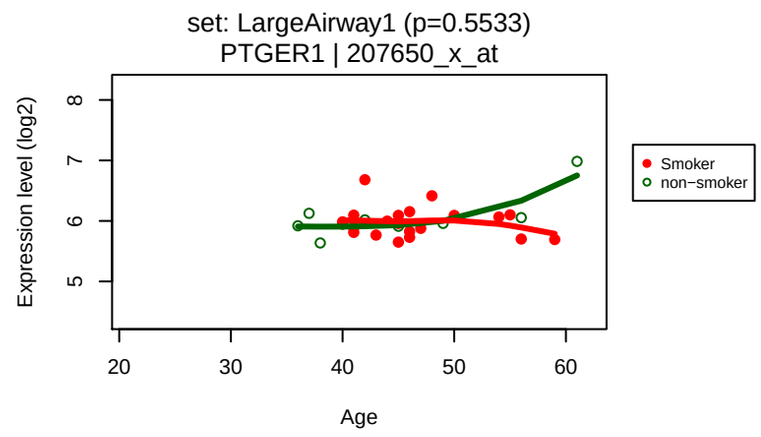

Gene expression of smokers and nonsmokers
gene: PTGER1 | 207650_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1148
p=0.2983
p=0.0127
p=0.5533
p=0.0591
p=0.979
p=0.6165
p=0.1643
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

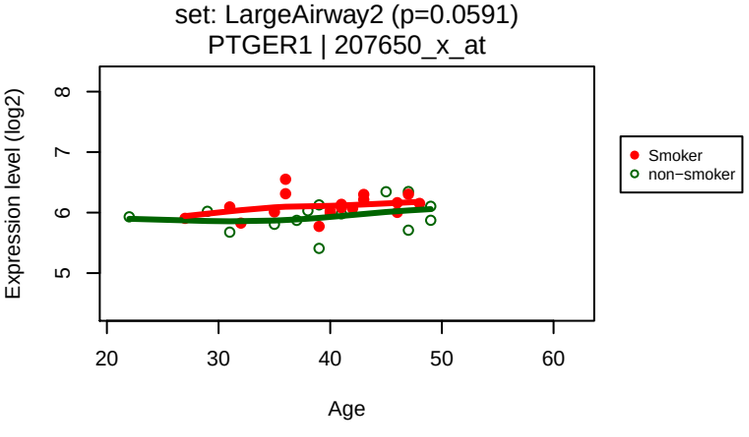

Gene expression ratio (smokers/nonsmokers)
gene: PTGER1 | 207650_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1148
p=0.2983
p=0.0127
p=0.5533
p=0.0591
p=0.979
p=0.6165
p=0.1643
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

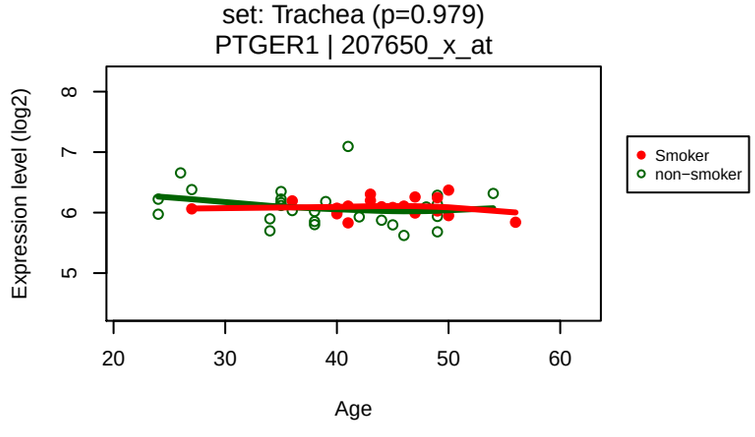

Gene expression ratio in Males
gene: PTGER1 | 207650_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8102
p=0.0871
p=0.7494
p=0.1393
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTGER1 | 207650_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3665
p=0.5336
p=0.874
p=0.7964
SmallAirway2
LargeAirway2
Trachea
Macrophage2

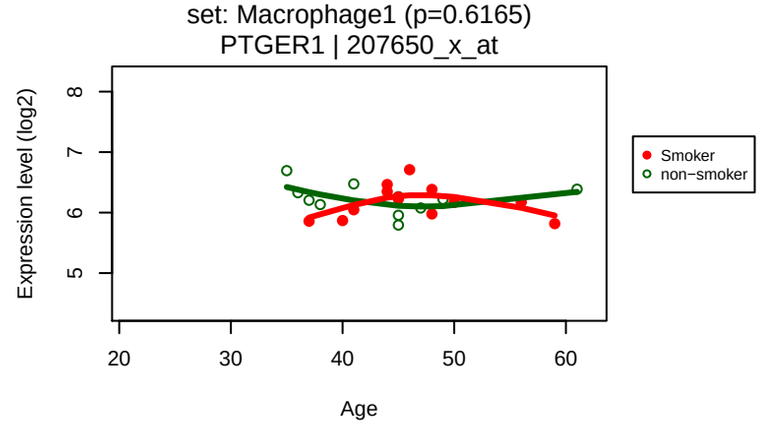

set: SmallAirway1 (p=0.1148)
PTGER1 | 207650_x_at
Expression level (log2)
Age
Smoker
non−smoker

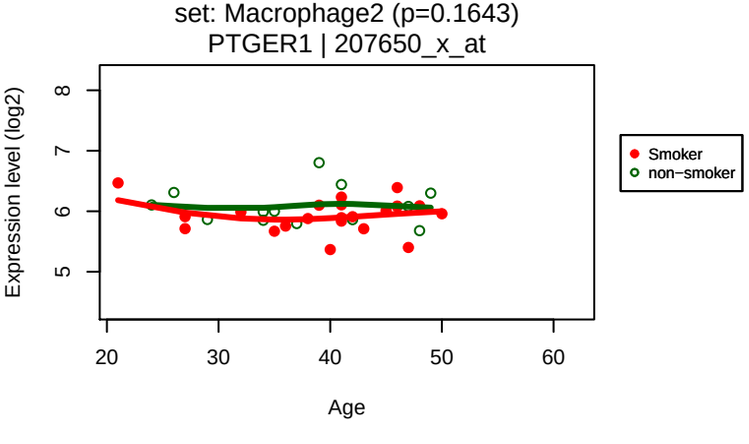

set: SmallAirway2 (p=0.2983)
PTGER1 | 207650_x_at
Expression level (log2)
Age
Smoker
non−smoker

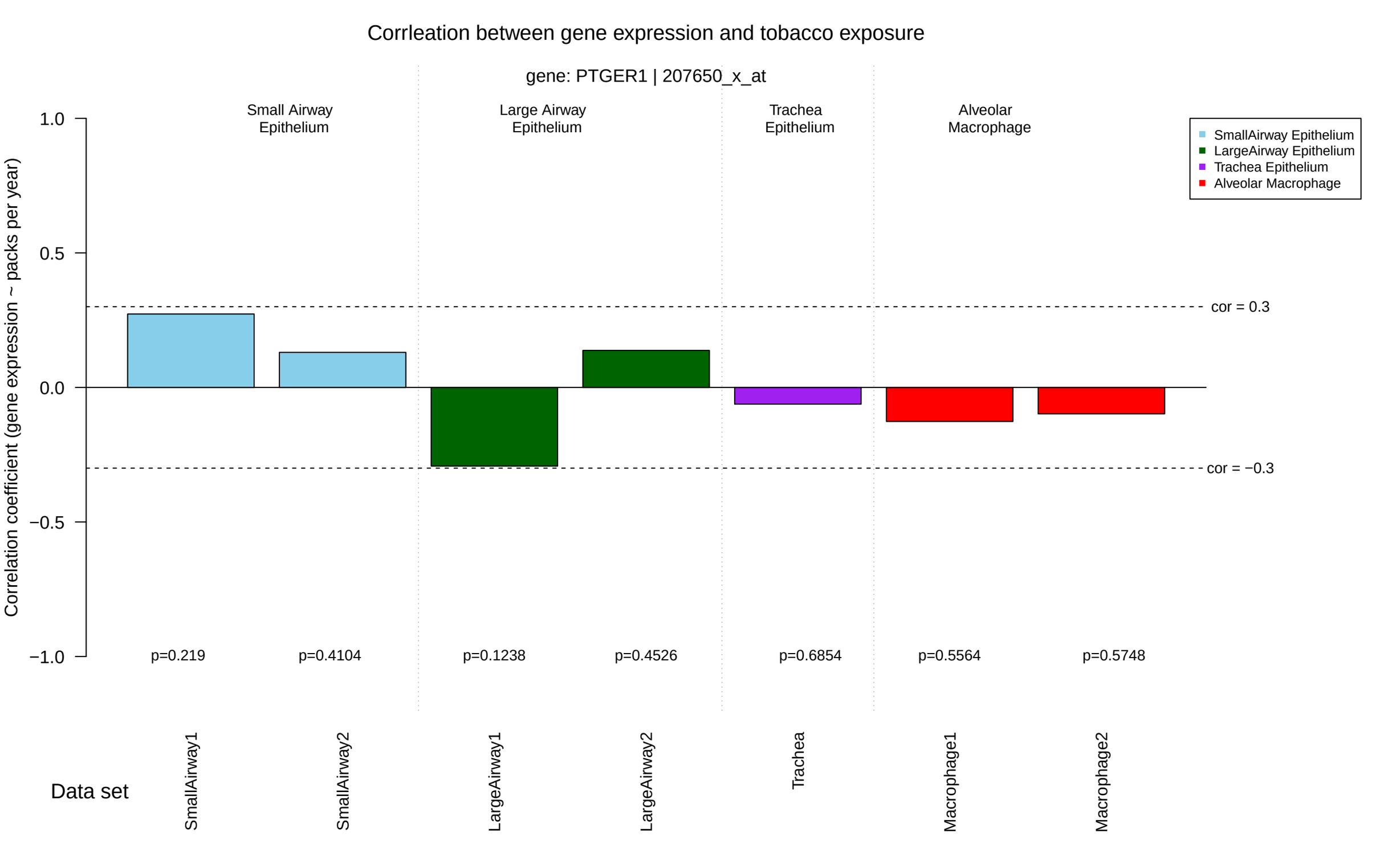

set: LargeAirway1 (p=0.5533)
PTGER1 | 207650_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0591)
PTGER1 | 207650_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.979)
PTGER1 | 207650_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6165)
PTGER1 | 207650_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1643)
PTGER1 | 207650_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTGER1 | 207650_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.219
p=0.4104
p=0.1238
p=0.4526
p=0.6854
p=0.5564
p=0.5748
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2